

Registration: Functional Connectivity Predicts Working Memory Performance -- A Multi-Site fMRI Study

OSF Registration: osf.io/abc123

Hypotheses

H1 (Primary): DLPFC-PPC functional connectivity strength will positively predict working memory accuracy (direction: greater).

Primary Outcomes

- O1: DLPFC-PPC functional connectivity (Fisher z-transformed), measured at baseline scan

Sample Size

- Planned N: 120 participants

MRI Acquisition Parameters

- Scanner field strength: 3.0 Tesla

Preprocessing Pipeline

fMRIPrep 23.1.0 with ICA-AROMA denoising. Cross-vendor robustness checks will be performed by comparing results between Siemens and GE scanners.

Uncertainty Quantification

Bootstrap confidence intervals (1000 resamples) with physics-informed signal-to-noise estimation.

Exclusion Criteria

- Head motion > 3mm translation or 3° rotation

Analysis Plan

SA1: Mixed-effects regression with WM_accuracy as DV, connectivity + scanner_vendor + age + sex as IVs, site + head_motion_fd as covariates. Software: FSL 6.0 + Python Nilearn.